OOS Number: OOS- 262173

Phase I

Section A – Test Details					
Initiator:	Ayomide Odugbesi (Written by: Olugbenga Ajayi)	**Test Date:**	14-Sep-2026	**Date Initiated:**	18-Sep-2026
Date of Incident:	18-Sep-2026	**Test Name:**	USP <71> / EP 2.6.1 Sterility Test	**Sample ID:**	ETX-260911-0654
Name of Analyst who Performed the Test:	Prepping & Processing Analyst: Ayomide Odugbesi (AOD) Reading Analyst: Elysse Nioupin (EN)	**Sample / Active Name**	ATROPINE SULFATE OPH SOLN 0.05% [2026]	**Lot #:**	09112026@3
		Dosage Form	Liquid		
Description of Incident:	On 18 Sep 2026, sample ETX-260911-0654 was found positive for viable microorganisms after USP <71> sterility testing.	☑ Client Care notified of OOS ☑ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS			
	SOP / Test Method #: MICRO-SOP-5	Effective Date: 24Jul2026	SOP / Test Method Rev: 19	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 18-Sep-2026	Quality Assurance Department notified:	Yes ☑ No ☐	

Section B – Investigation and Root Cause Analysis			
Personnel Discrepancy		**Comments**	
Analyst interviewed?	☑ Yes ☐ No ☐ N/A	Yes, analysts Elysse Nioupin, and Ayomide Odugbesi were comprehensively interviewed.	
Correct samples analyzed?	☑ Yes ☐ No ☐ N/A	Yes, sample ID: ETX-260911-0654	
Correct SOP / Test Method used/followed?	☑ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Correct sample collection/technique?	☑ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Are the Test Result Records Attached & Calculations Verified and Correct?	☑ Yes ☐ No ☐ N/A	Yes, Information is available on Eagletrax under ETX-260911-0654.	
Is the analyst qualified to perform the test?	☑ Yes ☐ No ☐ N/A	Yes, the analysts are trained and qualified by quality to perform the test	
Is formula worksheet provided and verified?	☐ Yes ☐ No ☑ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☑ N/A	Not Applicable	
Was the sample stored appropriately?	☑ Yes ☐ No ☐ N/A	Yes, the sample was stored as per client's instructions. Information is available in EagleTrax Sample Location History under ETX-260911-0654	
Material Discrepancy		**Lot Number**	**Expiry Date**
Is the integrity of the sample(s) questionable?	☐ Yes ☑ No ☐ N/A		
Was the sample transported appropriately?	☑ Yes ☐ No ☐ N/A		
Are the other test results from this run under investigation?	☐ Yes ☑ No ☐ N/A		

OOS Number: OOS- 262173

Appropriate/correct glassware used?	☐Yes ☐ No ☑N/A		
Are reagents stored in proper condition and not expired?	☑Yes ☐ No ☐N/A	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs
Standard Used	Not applicable	Not applicable	Not applicable
Solvent Used	Not applicable	Not applicable	Not applicable
Is the standard the source of the root cause?	☐Yes ☐ No ☑N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐Yes ☐ No ☑N/A		
Equipment Discrepancy		**Equipment ID**	**Calibration Due Date**
Was the instrument set up correctly?	☑Yes ☐ No ☐ N/A	ISO 5 BSC E001314	Jun 2027
Is the clean room facility certified?	☑Yes ☐ No ☐ N/A	E001737 (CR115)	Dec 2026
System Performance Check (if applicable)?	☐Yes ☐ No ☑ N/A	Not applicable	Not applicable
System Suitability RSD < 2% (PDA)?	☐Yes ☐ No ☑ N/A	Not applicable	Not applicable
System Suitability RSD < 10% (QDA, CAD)?	☐Yes ☐ No ☑ N/A	Not applicable	Not applicable
Do the environmental monitoring results conform to required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☑Yes ☐ No ☐ N/A	Incubator E001356 (Monitored by Sensor E001450) Incubator E001357 (Monitored by Sensor E001449) Incubator E001034 (Monitored by Sensor E001501) Incubator E001031 (Monitored by Sensor E001505)	Jan 2027 Feb 2027 Jan 2027 Feb 2027 Aug 2027 Feb 2027 Aug 2027 Feb 2027

OOS Number: OOS- 262173

Are chromatogram/spectra/raw data correct?	☐ Yes ☐ No ☑ N/A	Not applicable	Not applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

- ☑ N/A
- ☐ Re-injection of sample vial
- ☐ Re-injection of original standard vial
- ☐ Re-vialing of original working standard solution
- ☐ Re-vialing of original working sample solution
- ☐ Re-dilution of a working sample solution from the original stock
- ☐ Re-dilution of a working standard solution from the original stock standard
- ☐ Other: N/A DK 01Oct2026

Section D – Phase I Summary

Phase I Summary:

This investigation was performed as per MICRO-SOP-53 Sterility Test Out-of-Specification (OOS) Investigation Procedure.

All analysts involved in the prepping, processing, and reading of the sample – Ayomide Odugbesi, and Elysse Nioupin were interviewed comprehensively. Their answers are recorded throughout this document.

Upon arrival, the sample was stored in accordance with the Client's instructions. Analyst Ayomide Odugbesi verified the integrity of the sample throughout both the preparation and processing stages. No leaks or turbidity were observed at any point, verifying the integrity of the sample.

All reagents and supplies mentioned in the material section above were stored according to the suppliers' recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system.

The cleanroom suite (CR115) used for processing procedures comprises three interconnected sections: the innermost ISO 7 cleanroom (115B), which connects to the middle ISO 7 buffer room (115A), and then to the outermost ISO 8 anteroom (115). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 115B through 115A and into 115.

The ISO 5 BSC E001314, located in the innermost ISO 7 room, (115B), was used for sample processing steps. It was thoroughly cleaned and disinfected prior to each procedure in accordance with MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). Additionally, BSC E001314 was certified and approved by both the Engineering and Quality Assurance teams. Sample processing was conducted in the ISO 5 BSC E001314 in the innermost ISO 7 room, (115B) by Ayomide Odugbesi on 14 Sep 2026 as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).

On 14 Sep 2026, the sample vials for ETX-260911-0654 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each sample vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time.

The secondary disinfection happened in the ISO 8 anteroom (115), where the vials were again treated with acidified bleach and provided a 10- minutes contact time before processing. Subsequently, the vials were moved into innermost ISO 7 cleanroom (115B). Inside this cleanroom, the processing analyst, Ayomide Odugbesi, performed a final disinfection step, allowing an additional 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001314.

OOS Number: OOS- 262173

Phase I Summary (continued):

Inside the BSC, each container was aseptically opened, ensuring minimal exposure to the environment. The contents were then dispensed directly from the container into sterile canisters placed on a vacuum manifold for membrane filtration. The sample was processed and tested as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test). Media bottles were transferred into incubators E001356 and E001357 to begin incubation.

On 18 Sep 2026 (Day 4 of incubation), microbial growth was observed in one TSB (Tryptic Soy Broth) media bottle by analyst, Elysse Nioupin, and confirmed by Microbiology Lab Supervisor, Robin Seymour. The positive TSB bottle for ETX-260911-0654 was submitted for Differential Staining and Microbial Identification under ETX-260918-0256. Differential staining indicated the presence of Gram-positive rods. Definitive identification identified the isolates as Bacillus litoralis and Bacillus sp. (Gram (+) rods).

The culture media utilized were within their expiry period. Negative control media bottles for the membrane filtration method were handled, processed, and incubated in the same manner as the test samples. No microbial growth was observed in the corresponding negative controls.

Upon analyzing the environmental monitoring results, no microbial growth was detected on personnel monitoring plates, settling plates, or ISO 5 BSC surface sampling plates on the date of testing, the preceding testing date, or the subsequent testing date. Additionally, no microbial growth was recovered from the weekly active air sampling or the weekly surface sampling of anteroom, buffer room and cleanroom plates collected on 10 Sep 2026.

The absence of contamination on analyst glove plates and work-surface environmental monitoring plates collected on the date of testing indicates that there were no viable transfer of contaminants to the ISO 5 biological safety cabinet (BSC). Additionally, the lack of contamination in other samples processed during the same testing session further demonstrates that the testing environment was operating in optimal condition.

Therefore, considering that no viable organisms were recovered from the ISO 5 environment, personnel monitoring results were acceptable on the date of testing, and no microbial growth was recovered from the weekly active air and surface sampling of Cleanroom suite CR115, the cleanroom environment is not considered a credible source of the contamination detected in the test sample.

The analysts confirmed full compliance with cleaning procedures as outlined in MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 BSCs for Cleanroom suite CR115 was performed on 16 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). During both cleaning cycles, it was documented that all H_2O_2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom suite CR115.

Analyzing a 6-month sample history for Blue Door Pharmacies 71015 (E71015), this specific analyte “ATROPINE SULFATE OPH SOLN 0.05%” has had no prior failures using USP <71> / EP 2.6.1 Sterility Test during this period.

To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed on the same day. All other samples processed by the analyst and other analysts that day tested negative. These findings suggest that cross-contamination between samples is highly unlikely.

Based on the observations outlined above, it is unlikely that the failing results were due to reagents, supplies, the cleanroom environment, the process, or analyst involvement. Consequently, the possibility of laboratory error contributing to this failure is minimal and the original result is deemed to be valid.

OOS Number: OOS- 262173

Phase I Summary (continued):

N/A DK 10Oct26

OOS Number: OOS- 262173

Has root cause been identified as a possible laboratory error?
Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes
- ☐ If retesting is ***required***, complete Phase II of the OOS (initiate Form 3.100.019.F02).

☑ No, the OOS result is valid.

Reviewer comment(s):

Prepared by (*Print*)**:** Dhvanir Kansara	Signature/Date:
Lab Manager/Designee (*Print*)**:**	Signature/Date:

Section E – Quality Disposition

Can investigation be closed?

☐ Yes
- ☐ Original result is valid.

☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (*Print*)**:**	Signature/Date:

Table 1: Information for samples under investigation OOS-262173

Reading Analyst	Day of Incubation	Sample ID	Related Microbial ID	Media with microbial growth	Microbial Identification
Elysse Nioupin	04	ETX-260911-0654	ETX-260918-0256	1 x 100mL TSB	*Bacillus litoralis* *Bacillus sp* Gram (+) rods

Table 2: Environmental Monitoring for Analyst & Cleanroom

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMM YYYY)	Analyst (Initials)	Day /Week(s)	Observation	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing 14Sep2026								
Personnel (Left Touch and Right Touch)	Daily	09 Sep 2026	AOD	Date Before Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	14 Sep 2026	AOD	Date of Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	15 Sep 2026	AOD	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC) E001314								
Surface Sampling of ISO 5 (4 locations)	Daily	11 Sep 2026	GL	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	14 Sep 2026	AOD	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	15 Sep 2026	AOD	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	11 Sep 2026	GL	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	14 Sep 2026	AOD	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	15 Sep 2026	AOD	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling for CR 115 (E001737)								
Active Air Sampling of Cleanrooms	Weekly	10 Sep 2026	SMO	Week of Testing	No growth	N/A	N/A	None
Surface Sampling of Anteroom, Buffer room and Cleanroom for CR 115 (E001737)								
Surface Sampling of Cleanrooms	Weekly	10 Sep 2026	SMO	Week of Testing	No growth	N/A	N/A	None